

Designed by Remus Teodorescu 2026

LECTURE 5.1

Convolutional and Graph Networks

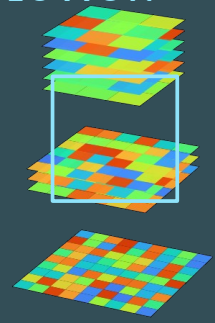
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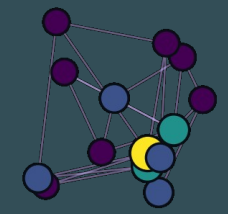
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Deep Learning for Engineering · 2026

CONVOLUTION · GRAPH NETWORKS



a grid — every pixel has the same neighbours



a graph — whoever the edges connect

A convolutional network reads an image through small windows and reuses the same kernel at every position, so its parameters do not grow with the image. A graph neural network does the same on a graph: every node updates from its neighbours with one shared rule, here on the six-bus network of the exercise.

Learning Objectives

SECTIONS, IN ORDER

1. **Local rules** — why a dense layer is a poor fit for an image
2. **Convolution** — what a convolutional network is, and its main pieces
3. **Graphs** — the same idea on data made of connected parts

One small example runs through the lecture, a six-bus power network. You work with it again in the exercises.

READ, AND WHAT EACH ANSWERS

UDL ch. 10.1	why is a dense layer a poor fit for an image?
UDL ch. 10	what does a convolution compute, and how is it set?
UDL ch. 13	how does the same idea work on a graph?

Recap

1 – 2 · DEEP NETWORK

1 What is a deep network, and what does depth give that width does not?

Shallow networks composed: the output of one layer is the input of the next. Each layer folds the input again, so the linear regions multiply with depth where width only adds them: more regions for the same number of parameters. *UDL ch. 4.1 – 4.5; Ex_04 notebook 04*

2 What made deep networks trainable, and how is a first one sized?

The gradient reaches the first layers as a product of many factors and can vanish on the way; ReLU and He initialisation keep it alive. Start at four layers of 32 neurons, and change the size only when the held-out error says so. *UDL ch. 7.5, 11; Ex_04 notebook 04*

3 – 4 · GENERALISATION

3 What is generalisation, and how do training and held-out error behave as capacity grows?

A low error on data the model has not seen. The training error falls all the way as capacity grows; the held-out error falls, then rises once the model starts to fit the noise. A fit through every point with a poor held-out error is an overfit. *UDL ch. 8.1 – 8.2; Ex_04 notebook 05*

4 How is a model chosen, and how is it kept from overfitting?

With three sets: train on the first, choose the capacity and the settings on the validation set, and report the test set once, at the end. Regularise before you shrink the network: weight decay and early stopping are the two methods to try first. *UDL ch. 8.5, 9.1; Ex_04 notebook 05*

Local Rules

Local rule — one small calculation that looks only at a point and its neighbours, with the same weights reused at every point. On a grid it is a convolution; on a graph it is message passing.

A DENSE LAYER ON AN IMAGE, AND WHAT REPLACES IT

- **a 256 × 256 image** — 65,536 inputs, one per pixel
- **a dense layer of 1,000 neurons** — about 65 million weights
- **the problem** — to it, the pixels have no neighbours
- **the new rule** — a small window, the same weights everywhere
- **one 3 × 3 kernel** — 10 parameters, for any image size
- **the lesson** — build in what we know about the data

ON A GRID — CONVOLUTION

computes	a weighted sum over a small window
parameters	the kernel weights and one bias
reused	at every position, so for any image size

ON A GRAPH — MESSAGE PASSING

computes	a sum over the connected nodes
parameters	one weight matrix and one bias
reused	at every node, so for any graph size

Convolutional Network

Convolutional network — a network for data laid out on a grid, such as an image. Each layer looks at small neighbourhoods and reuses the same weights at every position.

KIND, FUNCTION, ARCHITECTURE, APPLICATIONS

- **kind** — a feed-forward network whose layers are convolutions
- **function** — image in, features or label out, shift-equivariant
- **architecture** — convolution and pooling, then a dense head
- **applications** — classification, segmentation, detection, 1-D signals

AGAINST A DENSE NETWORK

	dense network	convolutional
connections	every input to every neuron	a small window only
weights	one per connection	one kernel, reused at every position
parameters	grow with the input size	weights and biases, any image size
a shifted input	an unrelated output	the same shift of the output
input size	fixed when designed	any, with global pooling

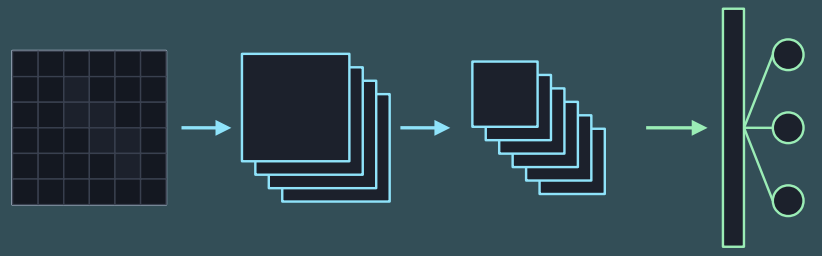
CONVOLUTION AND NETWORK

$$h_{ij} = a\left(\sum_m \sum_n k_{mn}\, x_{i+m,\, j+n} + b\right)$$

x the input image, k_{mn} the kernel, b one bias
a the activation, h_{ij} one output pixel

CONVOLUTION IN PYTORCH

```
conv = nn.Conv2d(1, 8, 3)  # eight 3 × 3 kernels
y = conv(x)                # same weights everywhere
```



input 16 × 16 × 1 | conv + pool 8 × 8 × 8 | conv + pool 4 × 4 × 16 | dense 3 classes

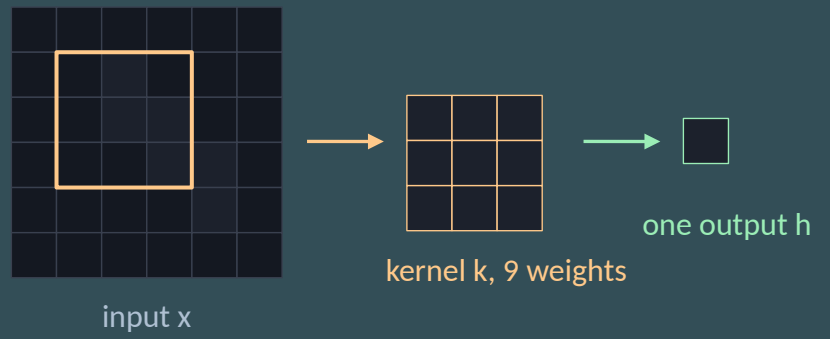
Kernel

Kernel — the small set of weights a convolution slides over its input. At each position it takes a weighted sum of the neighbouring values, adds a bias and applies the activation.

A WEIGHTED SUM OVER A SMALL WINDOW

- **a 3 × 3 window** — multiply, sum, add a bias, apply the activation
- **move one pixel and repeat** — with the same nine weights
- **the nine weights are learned** — once set by hand, now trained
- **the output** — an image of the same kind, one value per position
- **an edge detector** — weights −1, 0, +1 across the columns

ONE WINDOW, ONE OUTPUT PIXEL



THE SAME IDEA AS A WEIGHT MATRIX



Convolutional Network Hyperparametrization

Hyperparameters — the choices made before training that fix a convolution layer's shape: the kernel size, how far the kernel moves each step, how the border is filled, and how many feature maps go in and come out. The kernel weights are the parameters, learned in training.

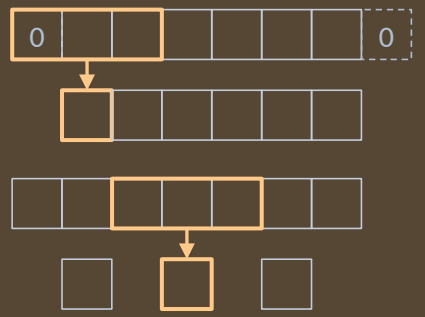
CHOSEN BEFORE TRAINING, THEN LEARNED

- **stride** — the step the window takes; a step of 2 halves the image
- **padding** — a border of zeros, so the output keeps the input's size
- **channels** — feature maps, each with its own kernel
- **parameters** — kernel area × maps in × maps out, plus biases

$$P_{\text{dense}} = N_{\text{in}} N_{\text{out}} + N_{\text{out}} \qquad P_{\text{conv}} = K^2 C_{\text{in}} C_{\text{out}} + C_{\text{out}}$$

$N_{\text{in}}, N_{\text{out}}$ neurons in and out, K the kernel side, $C_{\text{in}}, C_{\text{out}}$ channels in and out

STRIDE AND PADDING, IN 1D



stride 1, zeros at the edges
6 in, 6 out

stride 2, no padding
7 in, 3 out

HYPERPARAMETRIZATION IN PYTORCH

```
conv = nn.Conv2d(in_channels=1,   # grey image: 1 map in
                 out_channels=8,  # 8 kernels, 8 maps out
                 kernel_size=3,   # each kernel 3 x 3
                 stride=1,        # move one pixel a step
                 padding=1)       # zero border: 16x16 kept
pool = nn.MaxPool2d(2)            # halve each side: 16 -> 8
# learned: 3*3*1*8 weights + 8 biases = 80 parameters
```

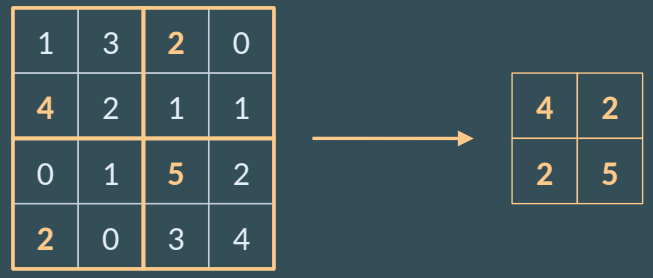
Pooling and Hierarchy

Pooling — a step that shrinks a feature map by summarising each small block, for example by its largest value. Stacked with convolutions, it builds features from edges up to whole objects.

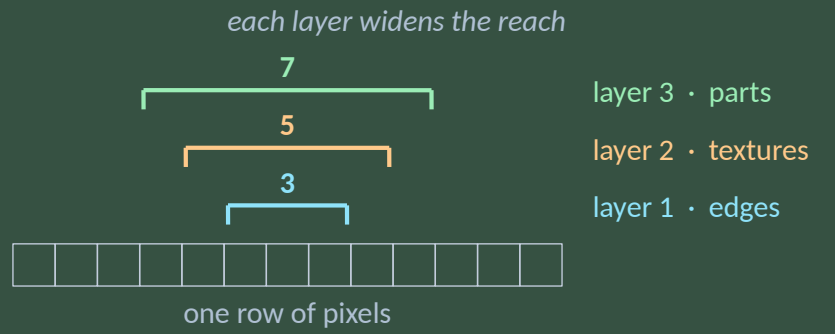
SHRINK THE IMAGE, BUILD UP FEATURES

- **pooling** — each 2 × 2 block becomes its largest value
- **why pool** — less computation; small shifts matter less
- **image shrinks, channels grow** — less where, more what
- **the hierarchy** — edges, textures, parts, whole objects
- **nobody designs it** — it emerges from training
- **when not to pool** — when the answer must say where

MAX POOLING, 2 × 2



WHAT ONE NEURON SEES, BY DEPTH



Equivariance

Equivariance — shift the input and the output shifts the same way. Invariance: shift the input and the output does not change. Convolution is equivariant; pooling adds invariance.

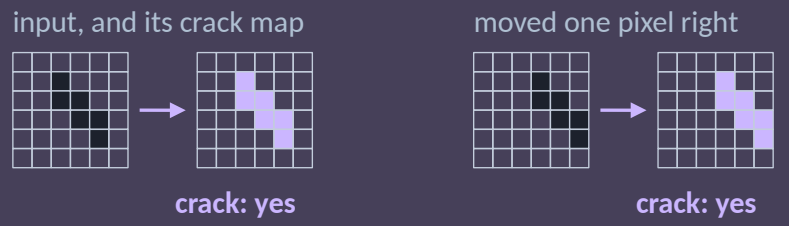
EQUIVARIANT AND INVARIANT

- **equivariant** — move the input, and the output moves with it
- **invariant** — move the input, and the answer stays the same
- **for example** — the crack map moves; “crack: yes” does not
- **convolution** — equivariant: the same kernel everywhere
- **pooling** — adds invariance by throwing position away

TALK · 2 MIN How do you understand convolution — what does it do to an image, in your own words? Then, in WHAT EACH OPERATION GIVES on the right, which row describes a crack detector that must say where the crack is, and which a classifier that only says whether there is one?

WHAT EACH OPERATION GIVES

	property	used for
convolution	equivariant	every layer
pooling	partly invariant	classification
global average	invariant	one label per image
no pooling	equivariant	segmentation



the map moves: equivariant · the answer stays: invariant

Convolution in Practice

A convolutional network in practice — a few convolution and pooling layers, then a dense head, trained on labelled images. In Ex05 notebook 01: weld radiographs of clean plate, cracks and pits.

WHAT THE CONVOLUTION SECTION SHOWED

- **local and shared** — small window, same weights everywhere
- **few parameters** — kernel weights and biases, for any image size
- **settings** — stride, padding and channels, chosen before training
- **pooling** — shrinks the image; deeper layers see more of it
- **equivariant** — output follows input; pooling adds invariance

IN EX05 NOTEBOOK 01



NOTEBOOK 01, THE RESULT

	parameters	train	held out
convolutional	2,019	1.000	0.989
dense	16,643	0.995	0.711

600 weld radiographs, 16 × 16 · clean, crack, pit

THE SAME RESULT IN UDL

convolutional	about 17 % test error
dense	about 40 % test error

MNIST-1D · same depth and width · structure built in wins

Graphs

Graph — a set of nodes joined by edges. The nodes carry values, the edges say which nodes are connected (and may carry values too), and there is no grid and no natural order.

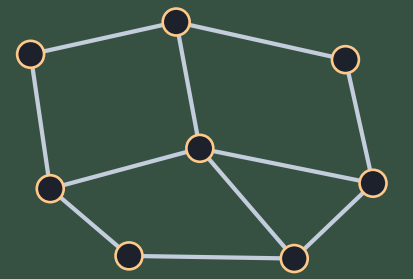
NODES AND EDGES, IN FOUR FIELDS

	nodes	edges
road map	junctions	roads between them
molecule	atoms	chemical bonds
electrical circuit	junctions	the components
power grid	buses	transmission lines

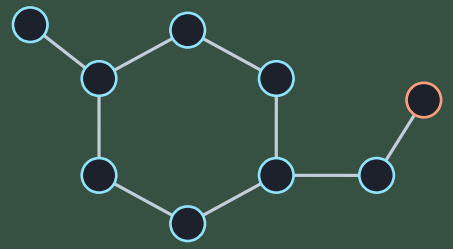
THE TYPES WE USE

- **undirected** — a line carries power both ways
- **directed** — one way only, like one paper citing another
- **edge features** — numbers on each edge, like line admittance

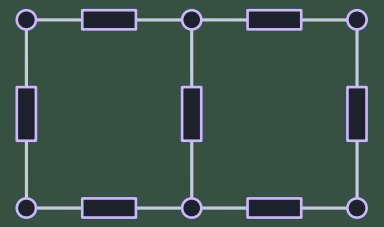
FOUR GRAPHS



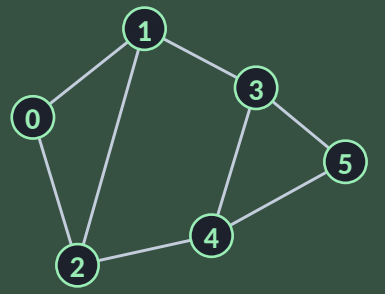
road map



molecule



electrical circuit



power grid, six buses

Graph Representation

Graph representation — the encoding of a graph for computation: typically an adjacency matrix or edge list defining the connectivity, together with a node feature matrix and, optionally, an edge feature matrix.

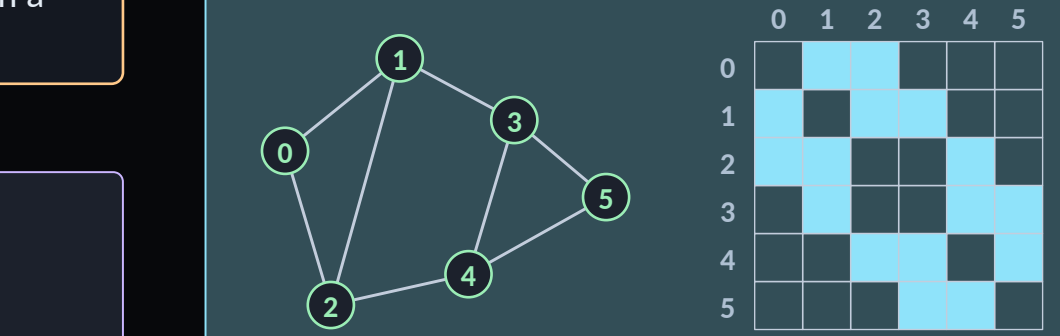
THREE MATRICES DESCRIBE THE WHOLE GRAPH

GRAPH REPRESENTATION IN PYTORCH

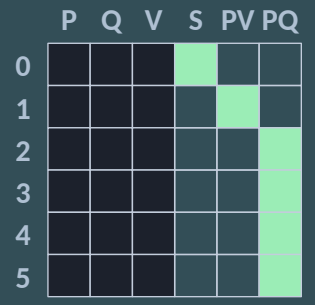
```
X = torch.tensor(bus_features)  # (6, 6): a row per bus
E = torch.tensor(line_RX)       # (8, 2): a row per line, R and X
A = torch.zeros(6, 6)           # adjacency, bus by bus
for i, j in lines:              # each of the 8 lines
    A[i, j] = A[j, i] = 1       # both ways: undirected
# sparse: 16 of the 36 entries of A are 1
```

TALK · 2 MIN How would you explain a graph to someone outside engineering, and what makes a power grid one? Then, in the adjacency matrix on the right, which row belongs to bus 2, and which three columns of it are lit?

THE SIX BUSES AS MATRICES



A, 6 × 6



X, 6 × 6

0–1	16.5
0–2	14.2
1–2	16.5
1–3	12.4
2–4	14.2
3–4	16.5
3–5	9.9
4–5	12.4

E, 8 × 1, |y|

Graph Neural Network

Graph neural network — its input is a graph. Each layer updates every node from its own value and its neighbours' (message passing). It is invariant to node count, numbering and topology changes.

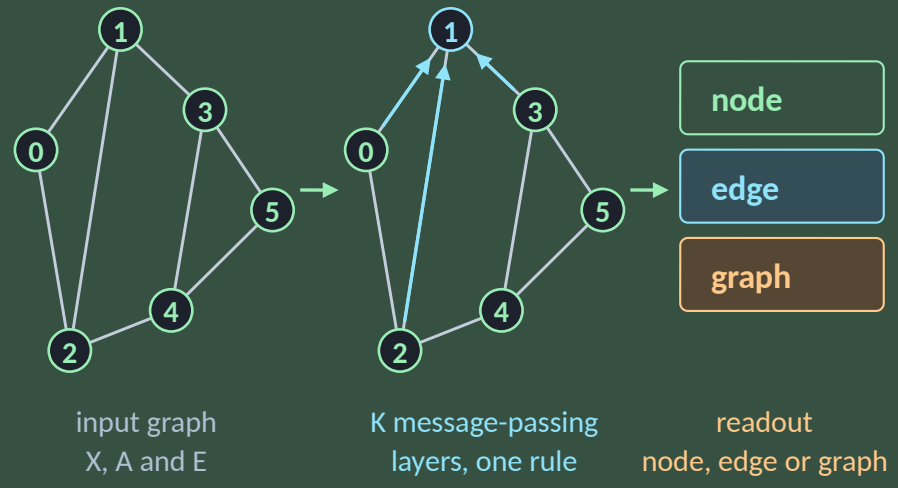
KIND, FUNCTION, ARCHITECTURE, APPLICATIONS

- **kind** — layers that update every node from its neighbours
- **function** — a graph in, X, A and E; an answer out
- **architecture** — K message-passing layers, then a readout
- **applications** — power grids, circuits, molecules, road maps

AGAINST A DENSE NETWORK

	dense network	graph network
input	one vector, fixed length	a graph of any size, with features
connections	every input to every neuron	along the edges only
weights	one per connection	one rule, shared by every node
renumbered nodes	an unrelated output	the same output, renumbered
topology	fixed when trained	an input; one model, many graphs

THE ARCHITECTURE, ON THE SIX BUSES



bus 1 gathers from buses 0, 2 and 3 · every bus does the same, with the same rule

Graph Tasks and Losses

Graph tasks and losses — a prediction for each node (bus voltage or angle), for each edge (line fault), or one for the whole graph (stability), trained with a regression or classification loss function as appropriate.

WHAT IS PREDICTED, AND HOW IT IS SCORED

task	output	loss
node regression	a number per node	squared error, as in notebook 03
node classification	a class per node	cross-entropy
edge prediction	a probability per edge	binary cross-entropy
graph classification	one class per graph	cross-entropy, on the averaged nodes

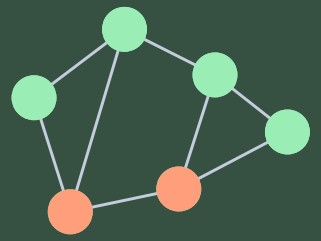
$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic} \log \hat{y}_{ic}$$

N nodes, C classes, $y_{ic} = 1$ for the true class, $\hat{y}_{ic}$ the predicted probability

GRAPH LOSSES IN PYTORCH

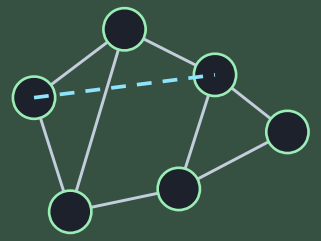
```
loss = F.mse_loss(out[:, 0], v)        # node regression: a voltage per bus
loss = F.cross_entropy(logits, labels) # a class per node, or per graph
g = out.mean(0)                        # a graph answer: average the nodes
```

THREE LEVELS OF ANSWER



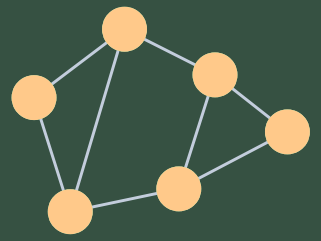
node

a value at every bus, as in notebook 03



edge

a probability for each line, such as whether it is present



graph

one answer for the whole grid, such as whether it is stable

Message Passing

Message passing — one layer of a graph network. Every node collects messages from its neighbours, aggregates them (sum, mean or max), and combines the result with its own state to get its new state. Each layer extends a node's reach by one hop, so the number of layers needed is related to the graph diameter, the longest shortest path between any two nodes.

HYPERPARAMETERS AND DIAMETER IN PYTORCH

```
A1 = torch.eye(6) + A              # a bus and its neighbours
R, hops = A1, 1                    # who has heard whom, after 1 hop
while (R == 0).any():              # someone not reached yet?
    R, hops = R @ A1, hops + 1     # one more hop
layers = hops                      # the diameter: 3 on the six buses
hidden, agg = 32, "sum"           # numbers per bus; sum, mean or max
act = nn.Tanh()                    # the activation in every layer
```

$$\mathbf{h}_i^{(l+1)} = \phi\left(\mathbf{h}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} \psi\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}\right)\right)$$

$\mathbf{h}_i^{(l)}$ node i at layer l, $\mathcal{N}(i)$ its neighbours, ψ the message, ϕ the combine

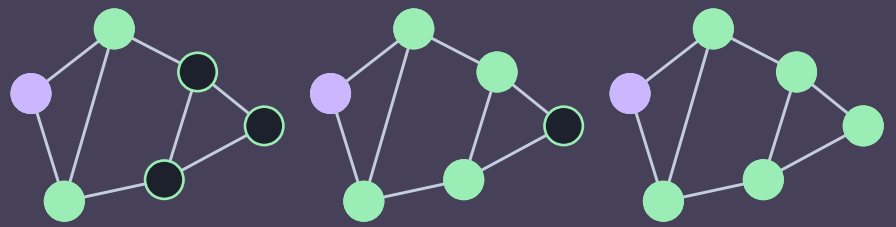
the physics sets the depth, weights the edges, gives the ground truth

ONE NODE'S UPDATE

ψ	Σ	φ
from each neighbour	the sum, order-free	with my own state

same functions at every node · repeat for more hops

HOW FAR BUS 0 HEARS, LAYER BY LAYER



layer 1: 3 of 6 buses | layer 2: 5 of 6 buses | layer 3: 6 of 6 buses

diameter 3: three layers reach every bus

Permutation Equivariance

Permutation equivariance — renumber the nodes and the node outputs are renumbered the same way; a graph-level output does not change at all (invariance). A GNN has this property by construction, because every node uses the same update and an order-independent aggregation.

RENUMBER THE NODES, THE OUTPUT FOLLOWS

- **bus numbers** — bookkeeping, with no natural order
- **a graph network** — the output follows exactly
- **a dense network** — no such guarantee

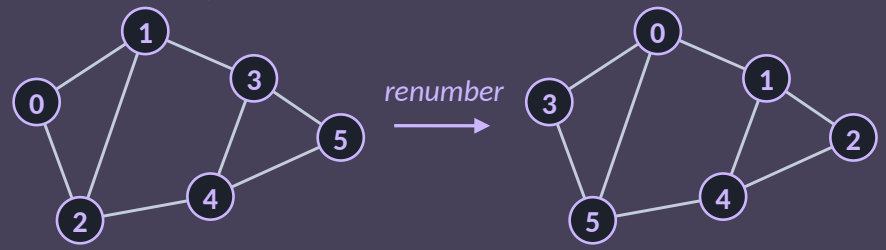
$$f(\mathbf{PX},\ \mathbf{PAP}^\top) = \mathbf{P}\, f(\mathbf{X}, \mathbf{A})$$

X node features, **A** the adjacency, **P** any relabelling of the nodes, f the network

RENUMBERING, CHECKED

```
perm = torch.randperm(6)                # renumber the six buses
out = gnn(x[perm], A[perm][:, perm])    # the same grid, relabelled
torch.allclose(out, gnn(x, A)[perm])    # True: the output follows
```

ONE GRID, TWO NUMBERINGS



same lines, same physics · only the labels changed

GRAPH AND DENSE, RENUMBERED

	graph network	dense network
renumber input	output renumbered	a different answer
because	one shared rule, a sum	a weight per position
input size	any number of buses	fixed when trained

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
residual networks	skips let very deep networks train	UDL ch. 11	image recognition at scale
batch normalisation	rescales outputs between layers	UDL ch. 11.4	very deep image networks
graph convolution	neighbours weighted by degree	UDL ch. 13.4	citation, social networks
graph attention	learns how much a neighbour counts	UDL ch. 13.8	molecules, recommendations
encoder–decoder	shrinks, then regrows the image	UDL ch. 10.5	medical image segmentation
inductive learning	predicts on graphs never seen	UDL ch. 13.6	new molecules, new networks

WHY THEY ARE LEFT OUT

Each of these earns its place on very large data sets: millions of photographs, molecules or social-network users. The problems in this course are small and physical, and today's convolution and message passing are enough for them.

Exercise 5

THE NOTEBOOKS, IN ORDER

- **01** — weld radiographs, clean, crack or pit: set kernels by hand, train a CNN, compare it with a dense network
- **02** — the six-bus grid as a graph: adjacency, neighbours, message passing, and the renumbering check
- **03** — a graph network learns the AC power flow: compared with a dense network, timed, and a line tripped
- **04** — after L5.2: persistence, a recurrent network and an LSTM forecast the hourly load
- **05** — the report: every notebook's answers, gathered, and one question to conclude

THE QUESTION EACH NOTEBOOK ANSWERS

01	can a CNN beat a dense network?
02	how does a node hear its neighbours?
03	how close to AC power flow, how fast?

REPORT QUESTION, TO CONCLUDE

Across a grid, a graph and a sequence: when did building the data's structure into the network beat a dense network, and what did it cost?

SUMMARY

Key Takeaways

CONVOLUTIONAL NETWORK	
what it is	a network for grid data: small kernels, shared everywhere
function	an image in; features, a label or a map out
train, tune	Adam on a loss; kernel size, stride, padding, channels
vs dense	far fewer parameters, and it generalised: 0.99 vs 0.71
used for	inspection images, 1-D signals, segmentation, detection

GRAPH NETWORK	
what it is	a network on a graph: nodes updated from their neighbours
function	a graph in; an answer per node, per edge or per graph
train, tune	Adam on a loss; layers from the diameter, width, aggregation
vs dense	as accurate here, any size, and it survives renumbering
used for	power grids, heating and gas networks, batteries, molecules

Next — L5.2: RNN and LSTM.

Questions

1 – 2 · CONVOLUTIONAL NETWORK

1 **What is a convolutional network, and what function does it implement?**

A network for data on a grid. A small kernel of shared weights is moved across the image and computes the same weighted sum at every position; stacked layers and pooling turn the image into features, then into a label or a map. *UDL ch. 10.1 – 10.4; Ex_05 notebook 01*

2 **How does it differ from a dense network, and why does it do better on images it has not seen?**

It has far fewer parameters, because one kernel is shared by all positions, and a feature learned in one place is found in every other: shifting the image shifts the output. In notebook 01 it scored 0.99 on unseen images against 0.71 for the dense network. *UDL ch. 10.2; Ex_05 notebook 01*

3 – 4 · GRAPH NETWORK

3 **What is a graph network, and what does one message-passing layer do?**

A network on nodes and edges. In each layer every node collects the features of its neighbours, combines them with its own through shared weights, and is updated. After k layers a node has heard from every node within k edges. *UDL ch. 13.1 – 13.4; Ex_05 notebook 02*

4 **How does a graph network differ from a dense network, and why does that suit a power grid?**

Its weights belong to the rule, not to the nodes, so the same network runs on a graph of any size and gives the same answer however the nodes are numbered. A tripped line is one edge removed, and nothing has to be trained again. *UDL ch. 13.2 – 13.6; Ex_05 notebook 03*

NEXT

L5.2 — RNN and LSTM

Today's structures were spatial, a grid and a graph. Next time the structure is temporal: how to carry information from the start of a long sequence to its end.

BEFORE THE NEXT SESSION

read	GBC ch. 10.1 – 10.2, 10.10
run	Ex_05 notebooks 00 to 03
bring	your tripped-line result
think about	layer count against graph diameter

both networks' errors, intact grid and after the trip, and the speed